

Beyond Semantic Confidence: Relation-Consistent Geometric Re-ranking for 3D Scene Graphs

Supplementary Material

Anonymous submission

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected by train/dev validation loss. Public preprocessing produces candidates for 533/548 contexts; the main evaluation retains the full 548-context ground-truth universe and treats the 15 contexts without candidates as empty sets. For sensitivity, the eligible-only denominator contains 3,899 relations, and a documented recovery covers all 548 contexts by lowering the visible-object minimum to two and regenerating views for two scans.

SGFN. We use the official `full_1160` source with its 160-object, 26-relation checkpoint on the same evaluation scans and denominator.

Guarantees and Compatibility Analyses

Proof of Proposition 1. For any $h \in H_a$, $C_a^{\text{tr}}(hx) = |H_a|^{-1} \sum_{g \in H_a} C_a(ghx)$. Right multiplication by h is a bijection on the finite group, so re-indexing $g' = gh$ gives $C_a^{\text{tr}}(hx) = C_a^{\text{tr}}(x)$. The two-element proximity and vertical actions yield the identities in the main paper, and the singleton support/contact action is the identity.

Family-sequence preservation. For each family a , the algorithm constructs the ordered list π_a defined in the main paper. At position j , it selects the next unselected candidate from π_{a_j} ; hence the output family sequence is exactly $(a_1, \dots, a_n)$. The support/contact list follows the source ranking, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. Therefore the support/contact selections and their order are identical at every reported K .

Prefix utility optimality. Fix a prefix length K , and let $n_a(K)$ be the number of positions assigned to family a in the source prefix. Every feasible ranking that preserves these family counts must select $n_a(K)$ candidates from each family. Because π_a is sorted by u_i , RelCompat3D selects the $n_a(K)$ highest-utility candidates for every proximity/vertical

family. If another feasible selection omits one of these candidates in favor of a lower-utility candidate from the same family, exchanging the two increases the summed within-prefix utility. The support/contact subsequence is fixed, so it does not affect this comparison.

Counterfactual target construction. Measured-family ground-truth relations provide positives. Negatives are generated within the same scene context and linked to their source positive; no source score, predictor identity, or evaluation candidate is used. Table 1 gives the complete construction policy.

Table 1: **Counterfactual-negative construction.** Normalized XY distance uses the mean object-pair XY diagonal. Zero overlap requires both directed projected-overlap ratios to be at most 10^{-9} .

Family	Intervention and acceptance rule	Relation to verifier
Proximity	Choose directed same-context pairs in descending normalized XY distance; require distance ≥ 2.5 and zero projected overlap.	Shares distance/overlap primitives and the 2.5 satisfied boundary; the verifier declares violation only at ≥ 3.5 .
Vertical	On the same directed pair, replace higher than with lower than or conversely; require the positive direction to have $ \Delta z \geq 0.25$ m and normalized $ \Delta z \geq 0.15$.	Shares both signed primitives and thresholds.
Support/contact	Replace the subject or object within context; reject known support/contact pairs in either direction and floor subjects; require any two of normalized XY distance ≥ 2.0 , zero overlap, and vertical gap ≥ 0.30 m.	Shares these OBB primitives and thresholds; point-subtype evidence used by the reported verifier is not used to generate negatives.

At most two negatives are emitted per positive, 200 per context-family, and three times the number of positives per family. These caps limit repeated easy negatives; they do not equalize family counts. The export contains 28,977 positives and 37,477 negatives. Total rows by family are 23,008 proximity, 5,488 vertical, and 37,958 support/contact. The explicit overlap above is why verifier-derived V is reported as a

related diagnostic rather than independent physical-validity ground truth; the feature-removal analysis tests, but does not eliminate, this construct dependence.

RelCompat3D fits one logistic model per measured relation family. Numeric features use means and standard deviations estimated from the training split; missing values use the corresponding training mean. Inputs are predicate/family indicators T , predicate-independent pair-geometry measurements G , and explicit $T \times G$ interactions. The source relation score, source rank, predictor identity, and object class are excluded. Transformation-consistent augmentation preserves the proximity predicate under endpoint swap; vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

The 60,208 training rows yield 33,961 linked pairs, and transformation-consistent augmentation produces 86,032 optimization rows. We use 800 deterministic full-batch steps at learning rate 0.2; the three family models contain 22–24 parameters. The support/contact model is used only in the all-family comparison, while the RelCompat3D ranking rule preserves that family’s source ordering.

Computational cost. We benchmark the fixed CPU implementation with one process and one thread on an Intel Core Ultra 7 265KF. Timing begins from preloaded relation candidates with precomputed pair-geometry measurements and includes proximity/vertical compatibility, transformation averaging, family-specific list sorting, and output assembly. It excludes source-predictor inference, reconstruction and geometry association, file parsing, metrics, and bootstrap, so it is not end-to-end pipeline latency. Table 2 reports medians over five runs after one warm-up.

Table 2: **CPU re-ranking cost.** Open3DSG has 533 nonempty public-prediction contexts; its 15 empty official contexts require no scoring or sorting.

Predictor	Contexts	Scored candidates	Total (s)	ms/context
VL-SAT	548	110,424	2.492	4.548
Open3DSG	533	79,722	1.828	3.430
SGFN	548	110,424	2.512	4.584

The three stored heads contain 69 parameters; the primary proximity/vertical path evaluates 45 of them and adds no fitted fusion parameter. Peak process RSS, including the preloaded candidates, is 366.9 MiB. All five runs preserve the exact family sequence and support/contact order.

Table 3: **Training and development discrimination.** Labels comprise GT-positive/counterfactual pairs, not human physical-validity judgments.

Family	Train	Dev	Param.	Train pairs	AUROC	Brier
Proximity	20,902	2,106	22	9,562	1.0000	.0009
Vertical order	4,922	566	23	1,644	.9982	.0079
Support/contact	34,384	3,574	24	22,755	.9933	.0256

Across all 6,246 development rows, AUROC/AUPRC/Brier are .9973/.9973/.0157. On 3,516 linked development pairs, the positive wins 99.23% of the time with mean logit margin 14.13. Correct-predicate compatibility wins on 97.40% of 384 vertical positives. Exact transformation averaging gives a maximum compatibility difference of zero under proximity endpoint swap and under joint endpoint swap and inverse-predicate transformation for vertical order, on development and evaluation rows from all three predictors.

The factor ablation uses the same split and estimator class. At development, T -only, predicate-independent G -only, additive $T + G$, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and mismatched-pair geometry reduce the intended reliability signal. These results verify factor use and the implemented transformation identities; they do not convert the constructed target into independent physical-validity ground truth.

Table 4: **Feature-removal analysis.** Cells are Recall / verifier V under the same family-aware ranking procedure. Exact removes the normalized scalar consumed by the verifier; family removes all directly related distance or vertical measurements; alternative retains only projected overlap for proximity and horizontal-distance/overlap context for vertical order.

Source	Condition	$K = 50$	$K = 100$
VL-SAT	Source score	.9272/.0268	.9635/.0476
	Full model	.9277/.0197	.9658/.0295
	Exact verifier scalar removed	.9280/.0198	.9660/.0300
	Related measurements removed	.9272/.0268	.9655/.0468
	Alternative evidence only	.9280/.0268	.9642/.0471
Open3DSG	Source score	.4043/.1387	.5111/.1242
	Full model	.4418/.0342	.5692/.0324
	Exact verifier scalar removed	.4383/.0342	.5692/.0325
	Related measurements removed	.4119/.0954	.5257/.0956
	Alternative evidence only	.4106/.1193	.5330/.1188
SGFN	Source score	.7402/.0385	.9235/.0630
	Full model	.7450/.0263	.9303/.0350
	Exact verifier scalar removed	.7465/.0268	.9303/.0357
	Related measurements removed	.7470/.0388	.9298/.0616
	Alternative evidence only	.7462/.0386	.9277/.0625

Every feature-removal model is refitted on the 1,061 training scans with the same linked-pair objective, exclusion of the source relation score, and exact transformation averaging. The exact-scalar condition retains development AUROC of .99999 for proximity and .99794 for vertical order. Removing all related distance or vertical measurements lowers these to .9512 and .7089; alternative evidence yields .8763 and .5531. At $K = 50$, the strict conditions retain a clear Open3DSG improvement but not the full V gain on VL-SAT or SGFN. At $K = 100$, all feature-removal conditions improve both point metrics across all three predictors. The results show that performance does not depend solely on a single verifier input, but they do not eliminate broader construct dependence on

correlated geometry.

Table 5: **All-family product comparison at $K = 100$.** Cells are Recall / verifier V. Unlike the primary family-aware ablation, compatibility is applied to every family; Open3DSG uses the documented full-coverage sensitivity so each corruption has a donor in every context.

Condition	VL-SAT	Open3DSG	SGFN
Structured product	.9688/.0325	.6055/.0339	.9418/.0372
Wrong predicate	.9507/.0745	.5753/.2315	.9182/.1120
Wrong-pair G	.9121/.0525	.2885/.0891	.8429/.0723
Shuffled G	.9089/.0615	.3326/.1418	.8338/.0861
Label-fixed swap	.9509/.0745	.5775/.2325	.9187/.1120
Distance only	.5554/.0981	.5038/.1071	.5554/.0981
Compatibility only	.6186/.0193	.5798/.0321	.6186/.0193

Counterfactual-Policy Sensitivity

We regenerate the training and internal-development targets and refit the proximity and vertical heads under nine one-factor-at-a-time conditions. Each variant recomputes normalization from the 1,061 training scans; no final-validation row enters target generation, normalization, or fitting. The default refit reproduces the main model and metrics exactly. Support/contact is kept in source order in every condition.

Table 6: **Counterfactual-policy sensitivity.** Order is linked-positive ordering accuracy over the re-ranked proximity and vertical sensitivity pairs on the 117-scan internal-development split. Source cells are Recall / verifier V. Only the named factor changes from the default.

(a) $K = 50$				
Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9277/.0197	.4418/.0342	.7450/.0263
Proximity 2.0	1.000	.9280/.0197	.4441/.0342	.7470/.0263
Proximity 3.0	1.000	.9275/.0197	.4423/.0342	.7437/.0264
Vertical 0.20 m	1.000	.9277/.0193	.4408/.0342	.7450/.0258
Vertical 0.30 m	1.000	.9277/.0204	.4439/.0342	.7447/.0274
Negative cap 1	1.000	.9277/.0197	.4431/.0342	.7447/.0263
Negative cap 4	1.000	.9277/.0197	.4418/.0342	.7455/.0263
Pair weight .125	1.000	.9277/.0198	.4421/.0342	.7450/.0264
Pair weight .5	1.000	.9277/.0197	.4418/.0342	.7450/.0262
(b) $K = 100$				
Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9658/.0295	.5692/.0324	.9303/.0350
Proximity 2.0	1.000	.9660/.0294	.5733/.0324	.9325/.0350
Proximity 3.0	1.000	.9653/.0296	.5662/.0324	.9277/.0350
Vertical 0.20 m	1.000	.9658/.0289	.5687/.0324	.9308/.0334
Vertical 0.30 m	1.000	.9658/.0306	.5680/.0324	.9300/.0370
Negative cap 1	1.000	.9658/.0295	.5697/.0324	.9293/.0350
Negative cap 4	1.000	.9658/.0295	.5680/.0324	.9305/.0350
Pair weight .125	1.000	.9658/.0296	.5695/.0324	.9303/.0351
Pair weight .5	1.000	.9658/.0294	.5687/.0324	.9303/.0347

Across all variants, the maximum absolute change from the default is .0023 Recall and .0011 V at $K = 50$, and .0040 Recall and .0020 V at $K = 100$. Every variant retains

lower V and no lower Recall than source scoring for all three predictors at $K = 50$ and $K = 100$. This reduces dependence on a particular threshold or negative count, but does not make the constructed target independent of the verifier.

Strong-Comparator and Open3DSG Sensitivities

Matched strong comparators. To separate score construction and model capacity, rank-average, RRF, and a small two-hidden-unit MLP use the same family-aware ranking procedure, support/contact source order, public/full 548-context target, and scan-cluster resampling as the RelCom-pat3D product. The MLP uses the same training rows, constructed target, feature contract, exclusion of the source relation score, and split as the linear heads; it is fitted once and applied to all three predictors without target-specific refitting.

Let q_i^Z and q_i^C denote descending within-context percentile ranks of the source relation score and compatibility, and let r_i^Z, r_i^C be their one-indexed ranks. Rank-average and fixed-constant Reciprocal Rank Fusion use

$$S_i^{\text{rank}} = \frac{1}{2}(q_i^Z + q_i^C), \quad S_i^{\text{RRF}} = \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \quad (1)$$

Table 7: **Matched fusion comparisons.** Cells are Recall / verifier V at $K = 50$ and $K = 100$. Every row preserves the source family sequence and support/contact selections.

Source	Fusion	$K = 50$	$K = 100$
VL-SAT	Product	.9277/.0197	.9658/.0295
	Matched MLP	.9272/.0189	.9650/.0296
	Rank-average	.9026/.0162	.9700/.0225
	RRF	.9174/.0180	.9710/.0247
Open3DSG	Product	.4418/.0342	.5692/.0324
	Matched MLP	.4670/.0413	.5989/.0371
	Rank-average	.4323/.0458	.5408/.0548
	RRF	.4237/.0948	.5476/.0772
SGFN	Product	.7450/.0263	.9303/.0350
	Matched MLP	.7457/.0258	.9288/.0350
	Rank-average	.7022/.0236	.9280/.0259
	RRF	.6813/.0253	.8774/.0302

At $K = 50$, the matched MLP has higher Open3DSG Recall than the product but also higher V; product and MLP are nearly tied on VL-SAT and SGFN. Rank fusion can lower V at large budgets but incurs larger predictor-dependent Recall losses. No fusion dominates the joint trade-off. A separate exact-label MLP uses stronger SGFN-specific correctness supervision and is therefore retained only as a source-specific upper comparator, not as an apples-to-apples baseline.

Family-conditioning ablation. The pooled product fits one compatibility model across all families and does not use the primary family-aware ranking procedure. Compared with Product (all families), it can increase Recall but generally raises V, particularly for Open3DSG and SGFN:

Table 8: **Family-specific versus pooled all-family products.** Cells are Recall / verifier V at $K = 50$ and $K = 100$.

Source	$K = 50$		$K = 100$	
	Family	Pooled	Family	Pooled
VL-SAT	.9293/.0203	.9300/.0219	.9688/.0325	.9690/.0387
Open3DSG	.4655/.0265	.4675/.0510	.6005/.0330	.6402/.0743
SGFN	.7706/.0256	.7925/.0292	.9418/.0372	.9413/.0464

Open3DSG coverage sensitivity. The public preprocessing drops a context when fewer than four annotated objects retain view metadata. This accounts for all 15 missing contexts. Table 9 distinguishes the eligible-context evaluation, the conservative full-target main evaluation, and a documented full-coverage recovery.

Table 9: **Open3DSG coverage sensitivity at $K = 100$.** Cells are Recall / verifier V. The main evaluation uses the full 3,972-relation denominator and no predictions for the 15 public-pipeline drops.

Evaluation	Contexts / GT	Source	RelCompat3D
Public eligible	533 / 3,899	.5206/.1242	.5799/.0324
Public, full target (main)	548 / 3,972	.5111/.1242	.5692/.0324
Recovered full coverage	548 / 3,972	.5161/.1242	.5743/.0332

The conclusion is stable to all three denominator and coverage choices. The main paper uses the conservative full-target public evaluation; recovered results are sensitivity evidence rather than the headline benchmark.

Scan-Cluster Confidence Intervals

The analysis resamples the 157 scan identifiers 1,000 times and carries every context of a sampled scan together. The ranking, model, and point estimates remain unchanged; shared bootstrap indices are used across conditions. A paired 548-context bootstrap gives the same directions and is retained as a finer-grained sensitivity.

All Recall point changes are non-negative. Recall intervals are strictly positive for Open3DSG at every K , VL-SAT at $K = 10/100$, and SGFN at $K = 50/100$; the remaining intervals overlap zero. Every Violation interval is strictly below zero except the tied SGFN $K = 5$ point and its $K = 10$ interval, which reaches zero. Thus the point-estimate trend is broad, while statistical support remains predictor- and budget-dependent.

External Transfer Stress Test

We apply the 3DSSG-trained compatibility model and family-aware ranking rule to the official ReplicaSSG test scenes using FROSS predictions. No target-specific parameter fitting, normalization, or imputation is performed in this evaluation. The exact mapping covers *near*, *above*, and *under*; support/contact is excluded because the ontologies do not provide an exact mapping. We report this as a transfer stress test rather than evidence of dataset-level generalization.

Table 11: **ReplicaSSG/FROSS transfer across all reported budgets.** R is exact-label Recall over 172 relations and V is verifier-derived Violation (lower is better). The experiment uses the same model and ranking rule as the main experiment.

K	Source R/V	RelCompat3D R/V	$\Delta R/\Delta V$
5	.0465/.0545	.0698/.0182	+.0233/−.0364
10	.0756/.0818	.1453/.0273	+.0698/−.0545
20	.1453/.1273	.2209/.0364	+.0756/−.0909
50	.2616/.1328	.3140/.0904	+.0523/−.0424
100	.3547/.1967	.3547/.1958	+.0000/−.0010

Paired scene-bootstrap intervals support joint improvement at $K = 10$ (ΔR [.0048, .1467], ΔV [−.1000, −.0091]) and $K = 50$ (ΔR [.0179, .0811], ΔV [−.0683, −.0185]). The $K = 20$ Recall interval touches zero, while $K = 5$ is inconclusive. At $K = 100$, heavy quantization of the source relation score makes multiplicative re-ranking nearly inert, and candidate support caps exact-label Recall at 44.19%. The stress test therefore supports budget-dependent transfer, not dataset-level generalization.

Family Composition within the Selected Top-100 Predictions

Table 12 reports each family’s slice inside the actual global top-100 ranking. Deltas compare RelCompat3D with the source relation score using shared paired context-bootstrap indices.

Table 12: **Family-specific metrics within the selected top-100 predictions.** Positive ΔV is a regression. Values are verifier-derived.

Source	Family	ΔR	ΔV
VL-SAT	Support/contact	.0000	.0000
VL-SAT	Proximity	+.0040	−.0076
VL-SAT	Vertical order	+.0051	−.0609
Open3DSG	Support/contact	.0000	.0000
Open3DSG	Proximity	+.0968	−.0443
Open3DSG	Vertical order	+.1538	−.3014
SGFN	Support/contact	.0000	.0000
SGFN	Proximity	+.0142	−.0150
SGFN	Vertical order	+.0051	−.0643

Support/contact is exactly unchanged because the ranking procedure preserves both its selection and the source family sequence at every K . Vertical order drives most of the aggregate reduction. The result is consequently a scoped operating-point improvement, not a uniform family claim. The same direction holds across the reported budgets; full machine-readable family slices and simultaneous intervals accompany the code artifact.

Verifier-Uncertainty Sensitivity

The reported $V@K_{all}$ includes uncertain candidates in its denominator but does not label them satisfied. We therefore

Table 10: **Scan-cluster intervals across all reported budgets.** Entries are RelCompat3D minus source relation score in percentage points with paired 95% intervals.

Predictor	K	ΔRecall	95% CI	$\Delta\text{Violation}$	95% CI
VL-SAT	5	+0.13	[0.00, 0.25]	-0.15	[-0.30, -0.03]
	10	+0.18	[0.05, 0.34]	-0.26	[-0.45, -0.10]
	20	+0.08	[-0.03, 0.19]	-0.28	[-0.44, -0.15]
	50	+0.05	[-0.05, 0.19]	-0.70	[-0.88, -0.55]
	100	+0.23	[0.05, 0.49]	-1.82	[-1.99, -1.66]
Open3DSG	5	+0.30	[0.12, 0.52]	-51.11	[-54.38, -48.14]
	10	+1.49	[1.01, 1.97]	-30.56	[-32.71, -28.43]
	20	+3.73	[3.01, 4.50]	-17.86	[-19.03, -16.66]
	50	+3.75	[2.96, 4.49]	-10.45	[-11.05, -9.86]
	100	+5.82	[4.70, 6.95]	-9.17	[-9.63, -8.73]
SGFN	5	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
	10	0.00	[0.00, 0.00]	-0.02	[-0.06, 0.00]
	20	+0.03	[-0.08, 0.14]	-0.25	[-0.44, -0.11]
	50	+0.48	[0.28, 0.70]	-1.22	[-1.42, -1.02]
	100	+0.68	[0.43, 0.98]	-2.80	[-3.04, -2.57]

recompute each ranking with $V@K_{dec}$ over satisfied/violated candidates, uncertainty rate $U@K$, and a pessimistic bound $V@K_{pess}$ that counts every uncertain candidate as a violation. Table 13 gives the $K = 100$ comparison.

Table 13: **Uncertainty sensitivity at $K = 100$.** Lower is better for all four reported quantities.

Source	Condition	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source score	.0476	.0729	.3464	.3941
VL-SAT	RelCompat3D	.0295	.0452	.3473	.3768
Open3DSG	Source score	.1242	.2126	.4161	.5402
Open3DSG	RelCompat3D	.0324	.0548	.4084	.4408
SGFN	Source score	.0630	.1005	.3732	.4362
SGFN	RelCompat3D	.0350	.0565	.3803	.4153

RelCompat3D-minus-source changes in V_{dec} are $-.0277$ for VL-SAT, $-.1578$ for Open3DSG, and $-.0440$ for SGFN; their paired 95% intervals all exclude zero. Uncertainty-rate changes are $+.0009$, $-.0077$, and $+.0071$, whereas pessimistic-V changes are $-.0173$, $-.0994$, and $-.0209$. Thus both decidable-only and pessimistic V are lower for all three predictors even though uncertainty does not decrease uniformly. The hard-rule baseline has zero Violation when the denominator includes all verifier statuses, but it retains uncertain candidates and therefore has pessimistic V of $.3712/.4722/.4042$. Thus zero measured violation does not imply reliable or complete selection.